Shuyu Gan

Autoresearch · Test-Time Reasoning · Multimodal Reasoning · LLM Post-Training (RL) · GUI Agents

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Education

University of Minnesota, Twin Cities Sep. 2025 – Present
Ph.D. Student in Computer Science
Advisors: **Dongyeop Kang** and **Mingyi Hong**

Zhejiang University Sep. 2020 – Jun. 2024
Bachelor of Engineering with **Honors** in Computer Science, Chu Kochen Honors College
GPA: **3.94/4.00**

Industry Research Experience

Microsoft, Applied Sciences Group Jun. 2026 – Sep. 2026
Research Intern *Mentor: Yinheng Li*

- Built TrainingDxBench, a benchmark in which an agent must diagnose and repair silent defects in a completed training run, toward autonomous training supervision for reliable autoresearch and recursive self-improvement (RSI); work in progress.

Zhipu AI Apr. 2025 – Jul. 2025
Research Intern *Mentor: Xiao Liu*

- Researched agentic RL for general, robust API-based computer-use agents (API-GUI paradigm: one line of Python per step, mixing GUI primitives with application-level API calls).
- Designed application-level API tools for desktop apps, expanded the task pool with verifiable evaluation functions, built behavior-cloning data from sampled trajectories, and took part in step-level GRPO training of the GLM-4-9B agent.

Simular Jun. 2024 – Nov. 2024
Research Intern *Mentor: Xin Eric Wang*

- Co-developed and open-sourced Agent S, a multimodal language-model agent combining experience-augmented hierarchical planning, continual memory updates, and an Agent-Computer Interface; built and maintained the codebase.
- Designed the ID-based Agent-Computer Interface, which tags every accessibility-tree element (plus OCR text blocks) with a unique integer ID for grounding actions without pixel coordinates; explored tree search and world-model look-ahead for action selection and integrated online web search into planning.
- Achieved state-of-the-art OSWorld performance at publication: 20.58% success versus 11.21% for the baseline (+9.37 percentage points; 83.6% relative improvement).

Preprints

SanSi: A Looped Typed Decision Model for System 1.5 Thinking
Shuyu Gan, Young-Jun Lee, Dongyeop Kang
arXiv:2610.07730, 2026. Under review

Publications

Do LLMs Recognize Your Latent Preferences? A Benchmark for Latent Information Discovery in Personalized Interaction
Ioannis Tsaknakis, Bingqing Song, **Shuyu Gan**, Dongyeop Kang, Alfredo Garcia, Gaowen Liu, Charles Fleming, Mingyi Hong
COLM 2026

Scaling Unverifiable Rewards: A Case Study on Visual Insights
Shuyu Gan, James Mooney, Pan Hao, Renxiang Wang, Mingyi Hong, Qianwen Wang, Dongyeop Kang
Findings of ACL 2026

A2P-Vis: An Analyzer-to-Presenter Agentic Pipeline for Visual Insights Generation and Reporting
Shuyu Gan, Renxiang Wang, James Mooney, Dongyeop Kang
IEEE VIS 2025 VISxGenAI Workshop; Honorable Mention

Agent S: An Open Agentic Framework that Uses Computers Like a Human
Saaket Agashe, Jiuzhou Han, **Shuyu Gan**, Jiachen Yang, Ang Li, Xin Eric Wang
ICLR 2025

Cultural Self-Adaptive Multimodal Gesture Generation Based on Multiple Culture Gesture Dataset
Jingyu Wu, Shi Chen, **Shuyu Gan**, Weijun Li, Changyuan Yang, Lingyun Sun
ACM Multimedia 2023

Academic Research Experience

SanSi: A Looped Typed Decision Model for System 1.5 Thinking Sep. 2026 – Present
Graduate Research Assistant, University of Minnesota *Advisor: Dongyeop Kang*

- Proposed SanSi, which turns a pre-trained looped LM (Ouro-1.4B) into a typed decision model that is read and trained (CE + Brier) after every loop, so one model serves every budget from one to eight loops in a single run.
- Built a suite of 10,027 test decisions from 59 sources; with the same data, recipe and seeds, SanSi reached 72.0% accuracy, 13.5 points above a same-shape non-looped model and 5.3 above a newer non-looped model of its size; SanSi-2.6B was 2.0 points above Qwen3.5-4B with 63% of its parameters.
- Showed that Ouro's frozen exit gate keeps the accuracy of eight loops with 44% fewer loops on average, and that loops extend the solvable depth beyond the training depths (72.6% on unseen liar chains, where a 3× larger single-pass model is at chance).
- Used as the only RL reward (no gold answers), SanSi raised the share of a generator's correct answers from 37% to 52%.

Selective Test-Time Scaling for Visual Insight Generation Jul. 2025 – Jan. 2026
Graduate Research Assistant, University of Minnesota *Advisor: Dongyeop Kang*

- Developed Selective Test-Time Scaling, a process-based refinement framework that allocates inference-time computation across pipeline stages and prunes weak branches using stage-specific evaluators.
- Built an end-to-end multi-agent data-analysis pipeline for profiling datasets, generating charts and visual insights, and producing reports.
- Evaluated LLM judges against expert preferences using stratified sampling and pairwise annotations; selected the best-aligned judge (Kendall's $\tau = 0.55$).
- In the main experiment, increased mean judge-rated report quality from 61.64 to 65.86 under a comparable total model-call budget.

Perceptual and Adversarial Guidance for Diffusion Transformer Training Jul. 2023 – Sep. 2023
Research Intern, mlPC Lab, University of California, San Diego *Advisor: Zhuowen Tu*

- Perceptual guidance: augmented latent Diffusion Transformer training (DiT-B/4, CelebA-HQ 256×256) with an image-space loss that decodes the predicted and ground-truth noisy latents through the VAE and matches their MLP-Mixer feature maps, blended with the standard noise-prediction MSE through a scheduled weight.
- Adversarial guidance: trained a timestep-conditioned ResNet-18 discriminator to separate predicted from real noisy samples (Diffusion-GAN style), alternating with DiT updates under a weighted GAN term and monitoring discriminator accuracy to keep the two players balanced.
- Reduced FID on CelebA-HQ from 13.77 (DiT baseline) to 11.63 at the best checkpoint.

Teaching and Service

Lead Teaching Assistant, CSCI 5541: Natural Language Processing, University of Minnesota Fall 2025
Reviewer: NeurIPS 2026 (ED Track); ICML 2026; ACL 2026 Workshop 2026

Awards and Honors

Outstanding Graduate of Zhejiang University 2024
Honorary Certificate, Chu Kochen Honors College, Zhejiang University 2024
First-Prize Scholarship and Outstanding Student, Zhejiang University (Top 2%) 2022–2023
Top-Notch Scholarship for Basic Disciplines, Zhejiang University (Top 5%) 2020–2022
Second Prize, National College Mathematics Competition 2021